INCH-POUND

MIL-M-28787/340
AMENDMENT 1
23 March 1994

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, ANALOG, KEY CODE EAP

This amendment forms a part of Military Specification MIL-M-28787/340(EC), dated 3 May 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4 and 5

TABLE I: Delete and substitute new table I as printed on pages 3 and 4 of this amendment.

PAGE 6

4.4.2, first line: Delete "A through M" and substitute "A through M and P".

4.5.1, third line: Delete "A through M" and substitute "A through M and P".

PAGE 7

FIGURE 2: Delete and substitute new figure 2 as printed on page 5 of this amendment.

PAGES 34 AND 35

Insert new pages 34 and 35 as printed on pages 6 and 7 of this amendment.

AMSC N/A FSC 5963
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(Project 5963-0257)

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TABLE I Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Continuous output current check, positive	II	S	60	72	60	72	60	72	60	72	60	72	mA
A3,A4	Continuous output current check, negative	II	S	-60	-72	-60	-72	-60	-72	-60	-72	-60	-72	mA
B1	Isolation													
	Initial	III	S	10	INF	10	INF	10	INF	10	INF	10	INF	MOhm
	End-of-life	III	S	1	INF	1	INF	1	INF	1	INF	1	INF	MOhm
B2,B3	Isolation													
	Initial	III	X	10	INF	10	INF	10	INF	10	INF	10	INF	MOhm
	End-of-life	III	X	1	INF	1	INF	1	INF	1	INF	1	INF	MOhm
C1,C2	Power supply current	IV	X	1	60	1	60	1	60	1	60	1	60	mA
D1,D2	Output offset voltage	V	X	-16.5	16.5	-16.5	16.5	-16.5	16.5	-16.5	16.5	-16.5	16.5	mV
E1,E2	Functional gain check	VI	X	0.95	1.01	0.95	1.01	0.95	1.01	0.95	1.01	0.95	1.01	V/V
F1,F2	Full power bandwidth check	VII	Q	4.77	INF	4.77	INF	4.77	INF	4.77	INF	4.77	INF	MHz
G1-G4	Output distortion	VIII	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	PCT
H1-H4	Resistance													
	Initial	IX	S	90	115	85	118	85	118	85	118	85	118	Ohm
	End-of-life	IX	S	85	118	80	120	80	120	80	120	80	120	Ohm
H5,H8	Resistance													
	Initial	IX	S	900	1150	850	1180	850	1180	850	1180	850	1180	Ohm
	End-of-life	IX	S	850	1180	800	1200	800	1200	800	1200	800	1200	Ohm

-See footnotes at end of table.

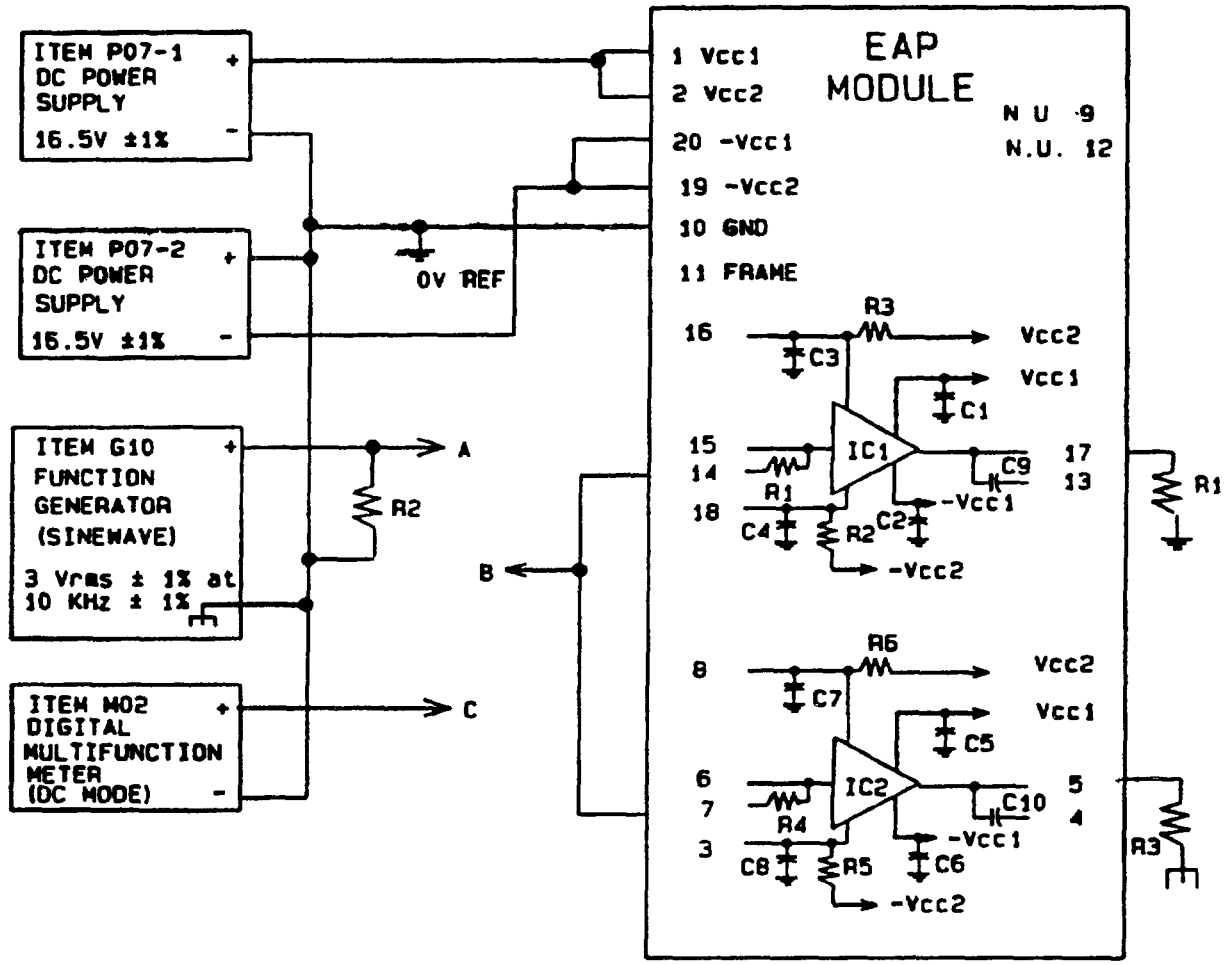
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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/ 3/	25°C±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C±5°C		60°C±5°C		-55°C±5°C		85°C±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
J1-J4	Power supply filter cap													
	Initial	X	S	0 18	0.22	0.15	0.25	0 15	0.25	0 15	0.25	0.15	0.25	µF
	End-of-life	X	S	0.15	0.25	0.15	0 25	0 15	0.25	0 15	0 25	0.15	0 25	µF
J5, J6	Capacitance													
	Initial	X	X	0.90	1.10	0.85	1.15	0 85	1.15	0.85	1.15	0.85	1 15	µF
	End-of-life	X	X	0 85	1.15	0.80	1.20	0 80	1 20	0 80	1 20	0 80	1 20	µF
K1, K2	Oscillation (capacitance load)	XI	Q	0	600	0	600	0	600	0	600	0	600	mVp-p
L	Continuity	XII	S	0	1	0	1	0	1	0	1	0	1	Ohm
M1-M4	Phase shift													
	Initial	XIII	S	-1	1	-1.5	1.5	1.5	1.5	-2	2	-2	2	Degree
	End-of-life	XIII	S	-1.5	1.5	-2	2	-2	2	-2 5	2.5	-2.5	2 5	Degree
N1, N2	Output latch													
	End-of-life	Fig. 2	Q	-15	15	-15	15	-15	15	-15	15	-15	15	mV
P	Power supply sequence	XIV	X	-1	10	-1	10	-1	10	-1	10	-1	10	mA

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life

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NOTES:
1. R1 = R3 = 1 KOhm ± 5 per cent, 1/4 W.
2. R2 = 50 Ohms ± 5 per cent, 1/4 W.
3. For life test, connect point B to pin 10 and for operating temperature and component temperature, connect point A to point B.
4. Following the life test period, prior to disconnecting the power, with point B connected to pin 10, connect point C to pin 17. The voltage shall meet the limits specified in table I for test condition N1.
5. For test condition N2, connect point C to pin 5.
6. Test condition N verifies that the amplifiers do not latch up.

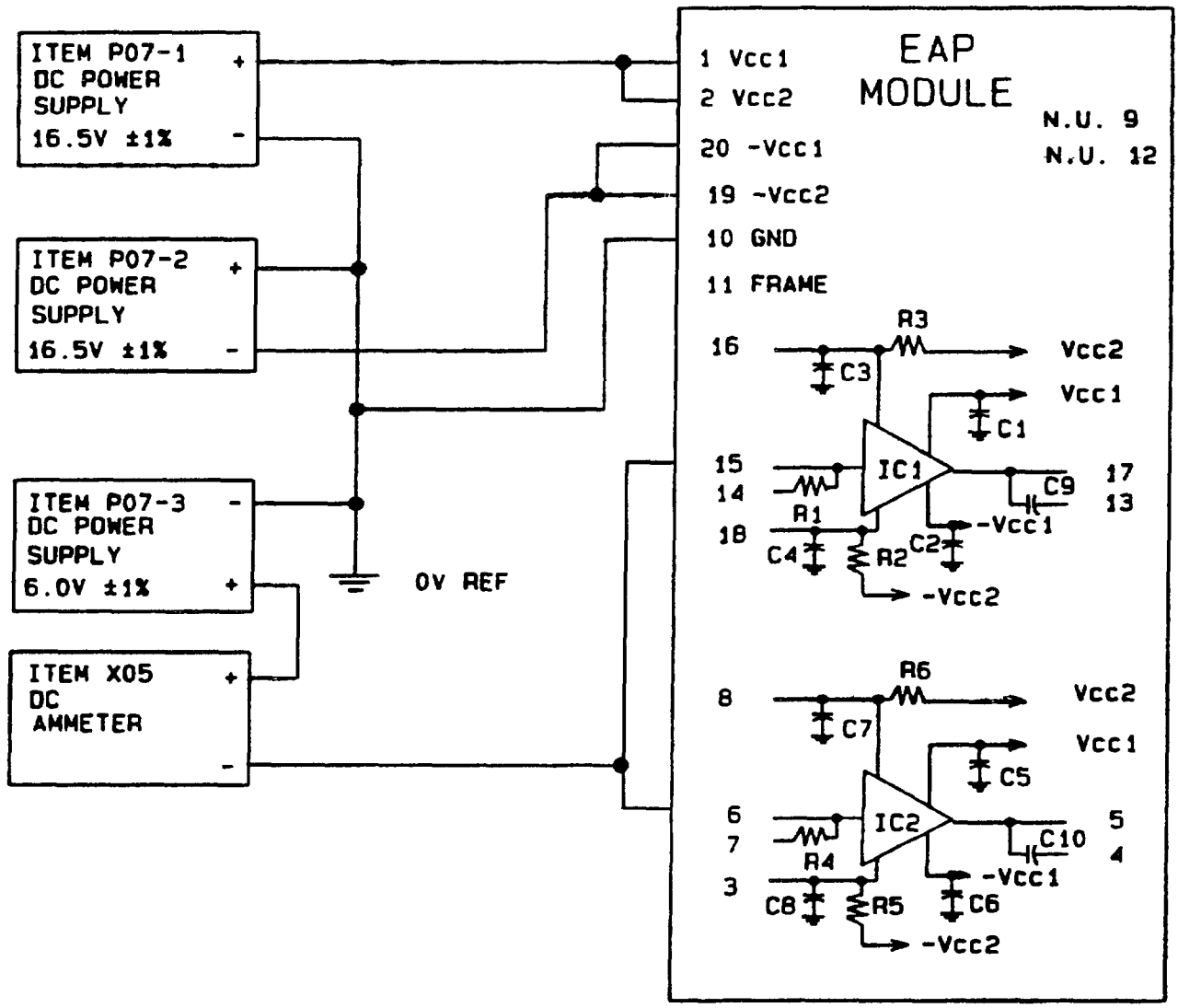
FIGURE 2. Life, operating, and component temperature test circuit.

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TABLE XIV. Power supply sequence test.

Test cond	Parameter	Initial limits 25°C ± 5°C ambient		Unit	Test figure
		Min	Max		
P	Power supply sequence	-1	10	mA	17

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NOTES:
Power supplies shall be turned on in the following order:
1. **P07-3**
2. **P07-2**
3. **P07-1**

FIGURE 17. Power supply sequence test circuit for table XIV.

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